

1. Sorting:-

Sorting is the process of arranging data in specific order, such as ascending or descending order, based on one or more values or attributes.

Ex:- Arranging students according to their marks.

2. Grouping

It is process of organizing data into categories or groups based on a common attribute. It helps in analyzing data within specific categories.

Ex:- Grouping students according to their department.

3. Ranking:-

It is the process of assigning a position or order to data items based on their values.

Ex:- Assigning Rank 1 to the student with the highest marks.

4. Plotting:-

It is the graphical representation of data using charts, graphs or visualizations. It helps in understanding patterns, trends in data.

Ex:- Represent student marks using a bar chart using code:-

```
import pandas as pd
import matplotlib.pyplot as plt
# create sample dataset
# create sample dataset
Name: ['Arjun', 'Bala', 'Chitra', 'Divya', 'Egha', 'Fathima']
Department: ['cse', 'ece', 'cse', 'ece', 'cse', 'ece']
Marks: [85, 78, 92, 88, 75, 95]
}
df = pd.DataFrame(data)
print('original data:')
print df
```


1. Sorting

print("Sorted by marks:")

```
sorted_df = df.sort_values(by='Marks', ascending=False)
print(sorted_df)
```

2. Grouping

print("Average Marks by department")

```
grouped_df = df.groupby('Department')['Marks'].mean()
print(grouped_df)
```

3. Ranking

```
df['Rank'] = df['Marks'].rank(ascending=False, method='dense')
```

print("\n|Students with Rank")

```
print(df.sort_values(by='Rank'))
```

4. Plotting

Bar chart of students' marks

```
plt.figure(figsize=(8,5))
```

```
plt.bar(df['Name'], df['Marks'])
```

```
plt.title('Student Marks')
```

```
plt.xlabel('Student Name')
```

```
plt.ylabel('rotation=45')
```

```
plt.tight_layout()
```

```
plt.show()
```

Plot average marks by department

```
grouped_df.plot(kind='bar')
```

```
plt.title('Average Marks by Department')
```

```
plt.xlabel('Department')
```

```
plt.ylabel('Average Marks')
```

```
plt.tight_layout()
```

```
plt.show()
```


Output

Name	Depr	Marks
1. Fathima	ECE	95
2. Chitra	CSE	92
3. Divya	ECE	88
4. Arun	CSE	85
5. Bala	ECE	78
6. Eghil	CSE	75

Average

CSE = 84

ECE = 87

Students With Rank

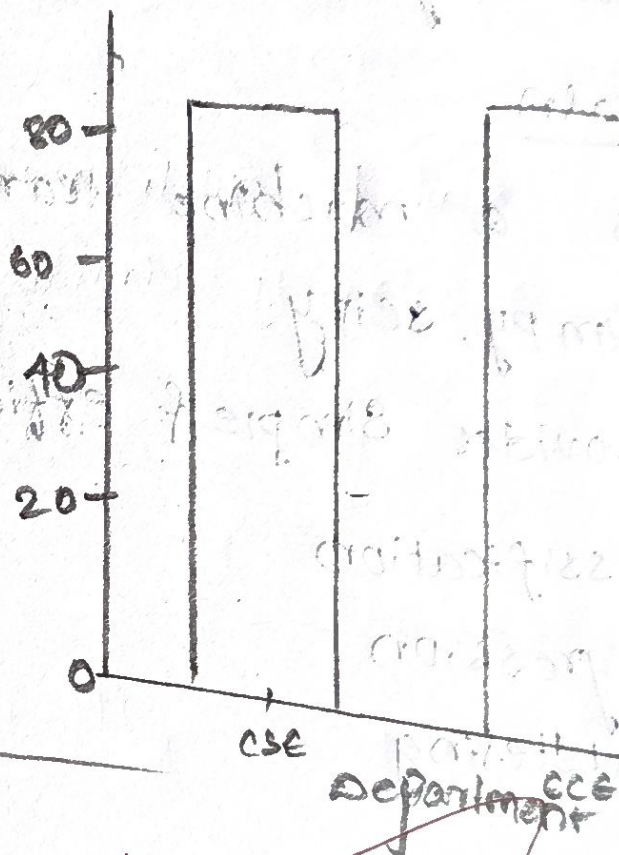
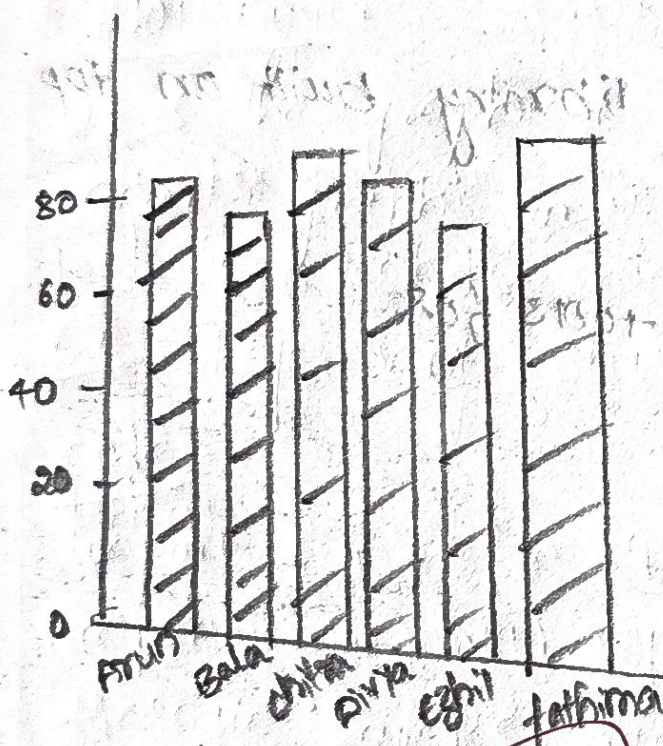
1. Fathima	ECE	95	1
2. Chitra	CSE	92	2
3. Divya	ECE	88	3
4. Arun	CSE	85	4
5. Bala	ECE	78	5

Vibin

0. Arun	CSE	85
1. Bala	ECE	78
2. Chitra	CSE	92
3. Divya	ECE	88
4. Eghil	CSE	75
5. Fathima	ECE	95

Sorted by marks

1. Fathima	ECE	95
2. Chitra	CSE	92
3. Divya	ECE	88



good

Vibin

23/7/26